# How Economists Discuss Central Bank Digital Currencies

## Themes, tone and stance, 2014–2026: a bibliometric and text-as-data study

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M2 – Private crypto

# The paper in one slide

**Research question.** How has economists and institutional discourse on CBDCs evolved between 2014 and 2026, and to what extent does this evolution align with changes in the policy environment?

| | |
|---|---|
| Who writes? | research economists, central banks and institutions |
| What do they write about? | themes and topics, and how they shift |
| How do they discuss CBDCs? | **tone and stance: the core of the paper** |
| Why, in their own words? | stated motivations |
| When, and with which environment? | bursts, policy events, sanctions |

**Scope.** Descriptive: we measure changes and show what moves with them; we do not claim causes.

48,188 texts collected $\rightarrow$ 17,709 with CBDC as a theme $\rightarrow$ 4,710 written by economists

# Outline

Introduction

Data

Who writes

What is written

How: tone and stance

Why: stated motivations

When: the environment

The discussion and its environment

Robustness

Conclusion

# Introduction

# Related literature: CBDCs

- Early CBDC research focused mainly on monetary policy, banking and financial stability (Barrdear & Kumhof, 2016; Bordo & Levin, 2017).
- The literature then turned to CBDC design: remuneration, account- vs token-based systems, accessibility and privacy.
- These design choices involve trade-offs between payment efficiency, financial inclusion and financial stability (Agur et al., 2019; Bindseil, 2020).
- More recently, the debate has broadened to cross-border payments, privacy and monetary sovereignty.
- **Our perspective:** rather than estimating one economic effect, we study how the economic and institutional debate itself has evolved.

# Related literature: what the field builds on

# What this paper adds to 29 earlier reviews

Five recent bibliometric reviews:

| Study | Data | Documents | Tools |
|---|---|---|---|
| Kvedaravičiūtė & Šapkauskienė (2024) | WoS + Scopus | 228 | VOSviewer |
| Fahad & Bulut (2024) | Scopus | 169 of 323 | Biblioshiny |
| Bapat & Sharma (2025) | Scopus | 371 | bibliometrics + TCCM |
| Prodan et al. (2024) | Scopus + WoS | 747 of 1,024 | PRISMA, VOSviewer |
| Paulovici (2026) | Scopus | 892 | VOSviewer |
| **This study** | **5 databases + central banks** | **4,710** | **+ text as data** |

- Gap: one or two databases, mostly English journal articles; central banks absent; who writes is not checked.
- Maps say *what* is studied, not *how* (tone, stance) or *why*; no measure is validated.
- Here: economists only (checked text by text), all languages, tone and stance over time, every coder validated.

Found in our own corpus: 29 titles announce a literature, systematic or bibliometric review of CBDCs (2019–2026).

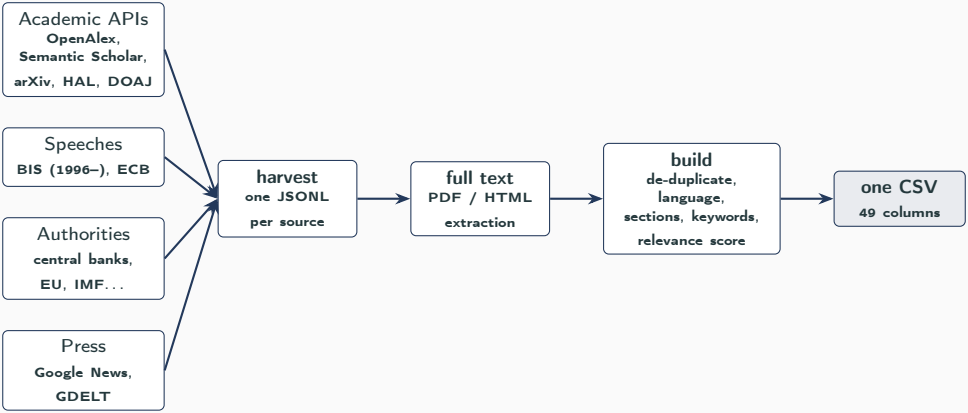
# Approach: one tool per question

| Question | Tool | Why this one | Explained |
|---|---|---|---|
| Who? | co-citation, co-authorship (association strength, Louvain); laws; log-odds with Dirichlet prior | standard science maps; log-odds shrinks rare words | slide 39, slide 42 |
| What? | thematic map; BERTopic (SBERT, UMAP, HDBSCAN); labels by c-TF-IDF | groups texts by meaning, across languages | slide 43, slide 44 |
| How? | tone: FinBERT on CBDC sentences ($\kappa = 0.52$); stance: reading + local LLM ($\kappa = 0.45$) | best of the coders tested against reading | slide 48, slide 49 |
| Why? | validated codebook ($\kappa = 0.46$) + covariate logit | beats SBERT, NLI, LLMs ($\kappa \leq 0.25$) | slide 47, slide 46 |
| When? | Kleinberg bursts; event studies with placebo dates; difference-in-differences | placebo fixes false positives (13%) | slide 50, slide 51, slide 53 |
| All | 7 robustness specifications; every coder checked on hand-coded texts | | slide 54 |

Every result slide links to the slide that explains its method (appendix).

# Data

# Pipeline




- 115 search phrases, 12 languages: *CBDC, digital euro, e-CNY, eNaira, Drex, digital fiat currency, Fedcoin, euro numérique, digitaler Euro, digital ruble...*
- Recall first: collect everything, then filter in four steps: homonyms, centrality, labels, economists.

# Step 1: is it really *Central Bank Digital Currency*?

```
TERMS = {
  "CBDC": r"\bCBDCs?\b|\bwCBDCs?\b|\brCBDCs?\b",
  "central bank digital currency":
      r"central[\s-]+banks?(?:['']s?)?[\s-]+"
      r"digital[\s-]+currenc(?:y|ies)",
  "digital euro": r"digital(?:e[rn]?)?\s+euro\b"
      r"|euro\s+numérique|euro\s+digitale?\b",
  "DCEP (e-CNY)": r"\bDC/?EP\b",
  ...
}
```

- "CBDC" is also a *skin disease*; "DCEP" a chemical.
- Ambiguous terms $t \in \mathcal{A}$ count only with monetary context:

$$\text{count}(t) = n_t \cdot 1\big[t \notin \mathcal{A} \vee \text{context}\big]$$

- context = *currency, payment, central bank, monnaie, Geld...*
- *digital dollar*, *Sand Dollar*: reported, never scored.

# Step 2: how central is CBDC to the text?

Each available field $f$ gets a centrality $c_f \in [0, 1]$:

$$c_{\text{title}} = 1[\text{CBDC term in title}] \qquad c_{\text{kw}} = 0.4 \cdot 1[\text{CBDC in keywords}]$$

$$c_{\text{abs}} = \frac{\#\{\text{sentences mentioning CBDC}\}}{\#\{\text{sentences}\}} \qquad c_{\text{body}} = \tfrac{1}{2} \min\left(1, \frac{\rho}{3}\right) + \tfrac{1}{2}\, \sigma$$

$\rho$ = mentions per 1,000 words; $\sigma$ = share of the ten slices of the text that mention CBDC.

$$\boxed{\text{score} = 100\left(0.6 \max_{f \in F} c_f + 0.4 \frac{1}{|F|} \sum_{f \in F} c_f\right)} \qquad F = \text{fields available}$$

One field clearly about CBDC suffices; a passing mention stays low; missing fields are not penalised.

# Step 3: labels, and how good is the filter?

| Label (all texts kept) | Texts | Theme | 95% CI |
|---|---|---|---|
| Main theme ($\geq 50$) | 15,464 | 29/30 | 0.83–0.99 |
| Secondary theme (25–49) | 2,245 | 11/29 | 0.23–0.56 |
| Mentioned only ($< 25$) | 489 | 1/30 | 0.01–0.17 |
| Press, text not available | 13,446 | 2/19 | 0.03–0.31 |
| Ambiguous terms only | 1,709 | 3/10 | 0.11–0.60 |
| Off-topic (homonyms, no match) | 14,803 | 0/20 | 0.00–0.16 |
| Dated before 2014 (checked: false) | 32 | – | – |

Theme = share read as having CBDC as a theme, in a stratified sample of 138 texts read blind to the label; Wilson intervals.

Kept: main + secondary theme (17,709 texts).

Strata $h$ weighted by size $N_h$; $T$ = theme labels:

$$\widehat{P} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h \in T} N_h}, \qquad \widehat{R} = \frac{\sum_{h \in T} N_h \hat{p}_h}{\sum_{h} N_h \hat{p}_h}$$

- Precision 0.89, recall 0.89.
- Secondary theme is noisy $\Rightarrow$ robustness on main theme only.

# Step 4: is it written by an economist?

| Texts with CBDC as a theme | Texts | |
|---|---|---|
| Research, classified in economics or finance (OpenAlex) | 3,783 | kept |
| Research, other field but an author at a central bank, IMF or BIS | 148 | kept |
| Research without OpenAlex record, economics venue | 63 | kept |
| Research in other fields | 3,469 | dropped |
| Central bank speeches | 511 | kept |
| Documents of central banks and international institutions | 205 | kept |
| Documents of governments and parliaments | 34 | dropped |
| Press articles (no author) | 9,496 | dropped |

Field = OpenAlex primary topic ("Economics, Econometrics and Finance"); institutions = authors' affiliations in OpenAlex. Research dropped: Computer Science 1,769, Business, Management and Accounting 663, no OpenAlex record 473, Social Sciences 281.

- 4,710 texts by economists: research economists, and central banks and international institutions (staff economists, institutional voice).
- Press dropped: no author is known, and journalists are not economists.

▷ method explained: slide 38

# Sources and completeness (economists' corpus)

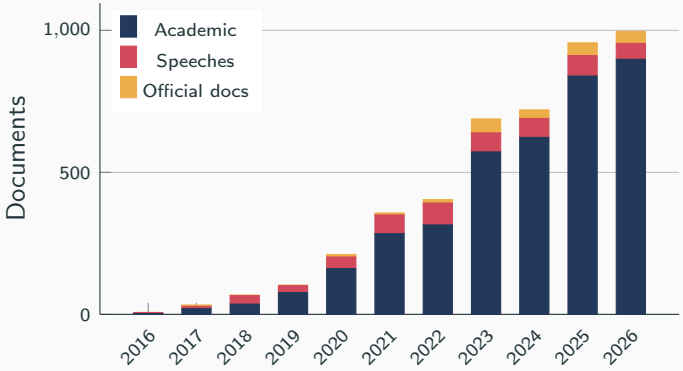
| Database | Documents |
|---|---|
| OpenAlex | 4,043 |
| Semantic Scholar | 2,164 |
| BIS central bankers' speeches | 465 |
| DOAJ | 251 |
| Central banks & authorities (crawl) | 205 |
| ECB speeches | 108 |
| HAL | 31 |
| arXiv | 24 |

Merged duplicates count in each database.

| Type of text | Documents |
|---|---|
| Academic | 3,994 |
| Speeches | 511 |
| Official docs | 205 |
| Total | 4,710 |

| Field available | Share |
|---|---|
| Exact day of publication | 71% |
| Issuer / outlet identified | 97% |
| Abstract | 68% |
| Full text | 63% |
| Keywords | 85% |
| Reference list | 32% |

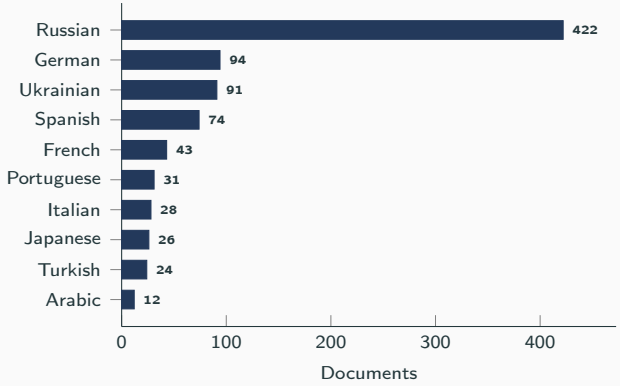
# A literature that took off after 2020




- From 103 in 2019 to 956 in 2025.
- Growth 60% a year.
- Turning point 2020: Libra (2019), COVID, the ECB report (Oct. 2020).

0 documents before 2016; 2026 runs to October.

# Languages




- English dominates: 3,721 texts (79%); the chart shows the others.
- Second language: Russian (422 texts).
- Everything kept in its original language: the topic model and the codebook are multilingual.

# Who writes

# Research economists: outlets, most cited, laws

**Main outlets**

| | |
|---|---|
| SSRN Electronic Journal | 461 |
| RePEc: Research Papers in Economics | 164 |
| Zenodo (CERN repository) | 126 |
| CyberLeninka (Russian repository) | 70 |
| Springer eBooks | 68 |
| Routledge eBooks | 56 |
| Ekonomika i Upravlenie | 47 |

**Bibliometric laws.** Growth 60%/year (2017–2025). Lotka: $f(n) \propto n^{-3.3}$, 83% of authors publish once. Bradford: 21 core sources hold 1/3 of the papers (zones 21 : 310 : 1,148).

**Most cited papers** (global citations)

| | | |
|---|---|---|
| Bordo (2017) | Central Bank Digital Currency and the Future of Monetary... | 517 |
| Auer (2020) | Rise of the Central Bank Digital Currencies: Drivers, App... | 454 |
| Agur (2019) | Designing Central Bank Digital Currencies | 433 |
| Andolfatto (2020) | Assessing the Impact of Central Bank Digital Currency on... | 404 |
| Barrdear (2016) | The Macroeconomics of Central Bank Issued Digital Currenc... | 341 |

Working papers (SSRN, RePEc) weigh as much as journals; a young, open field: most authors write once.

▷ method explained: slide 41

# Who speaks: central banks, institutions, research centres

**Speeches, by central bank**

| | |
|---|---|
| European Central Bank | 118 |
| Deutsche Bundesbank | 63 |
| Bank of France | 49 |
| Hong Kong Monetary Authority | 23 |
| Federal Reserve Board | 21 |
| Bank of England | 21 |
| Bank of Italy | 19 |
| Bank of Japan | 14 |

**Documents, by institution**

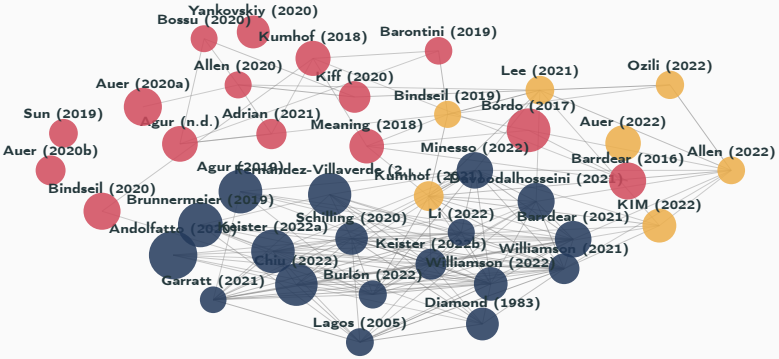
| | |
|---|---|
| European Central Bank | 98 |
| Bank of Japan | 27 |
| Sveriges Riksbank | 22 |
| Deutsche Bundesbank | 21 |
| Swiss National Bank | 9 |
| Central Bank of The Bahamas | 7 |
| Bank of Jamaica | 6 |
| Bank of Canada | 4 |

**Research, by author institution**

| | |
|---|---|
| Financial University | 63 |
| European Central Bank | 61 |
| International Monetary Fund | 39 |
| Bank of Canada | 38 |
| The Russian Presidential Acad... | 33 |
| Bank for International Settle... | 32 |
| Moscow State Institute of Int... | 29 |
| Lomonosov Moscow State Univer... | 29 |

European Central Bank is the most vocal central bank: the digital euro makes it the hub of the debate.

# Co-citation: the intellectual base




**Most cited within the corpus**

| | |
|---|---|
| Andolfatto (2020) | 264 |
| Brunnermeier (2019) | 218 |
| Bordo (2017) | 204 |
| Agur (2019) | 204 |
| Keister (2022) | 203 |
| Fernández-Villaverde (2020) | 199 |
| Chiu (2022) | 191 |
| Auer (2020) | 144 |

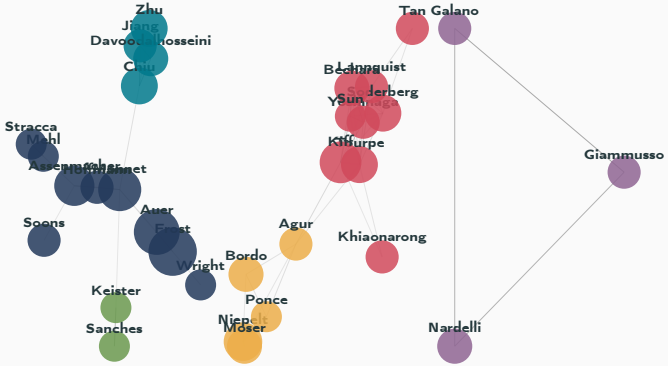
**Clusters** (largest members):

- Andolfatto (2020), Brunnermeier (2019), Agur (2019), Keister (2022a)
- Bordo (2017), Auer (2020a), Barrdear (2016), Bindseil (2020)
- Auer (2022), KIM (2022), Kumhof (2021), Lee (2021)

1,493 papers with reference lists, 13,810 distinct references; 40 most co-cited shown, 3 clusters, $Q = 0.10$. ▷ method

explained: slide 39

# Co-authorship: research communities




**Most productive authors**

| | |
|---|---|
| Ulrich Bindseil | 20 |
| Peterson K Ozili | 20 |
| Jon Frost | 16 |
| Annelieke Mooij | 16 |
| Raphael Auer | 14 |
| Cyril Monnet | 12 |
| John Kiff | 11 |
| Katrin Assenmacher | 10 |

5,716 authors; 48% of papers co-authored (mean team 1.9); 15% international. Top countries: RU 473, US 248, DE 167, CN 166, GB 153.

Papers with a central bank, BIS, IMF or World Bank author: 10% (23% in 2014–2019, 7% in 2023–2026), but 48% of citations.

▷ method explained: slide 39

# Who says what: research economists vs central bankers

Words most over-used by each group compared with the other ($z$-scores):

| **Research economists** (259,105 words) | | **Central bankers** (232,446 words) | |
|---|---|---|---|
| legal | 23 | payments | 28 |
| analysis | 21 | euro | 22 |
| development | 21 | european | 18 |
| research | 20 | europe | 18 |
| monetary | 20 | area | 18 |
| impact | 19 | services | 15 |
| model | 18 | innovation | 14 |
| implementation | 17 | conference | 14 |

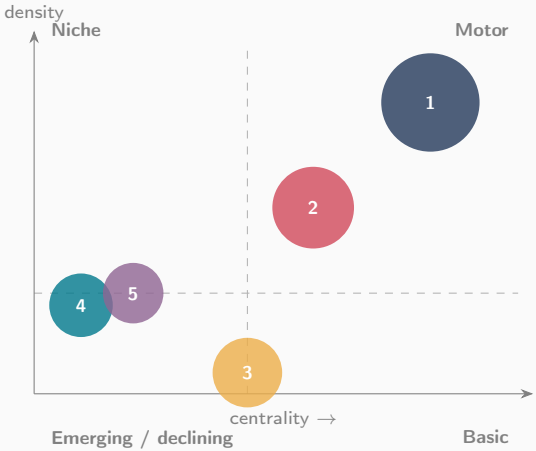
English texts; research: title + abstract; central bankers: speeches; topic words (*CBDC, digital, currency...*) removed.

Reading: research economists: *legal, analysis, development*; central bankers: *payments, euro, european*.

▷ method explained: slide 42

# What is written

# Thematic map: which themes drive the field?




Keyword clusters placed by centrality (links to other themes) and density (internal cohesion).

| | Theme (top keywords) | Quadrant |
|---|---|---|
| 1 | digital transformation in financial services, monetary policy, moneta... | motor |
| 2 | blockchain technology applications and security, digital transformati... | motor |
| 3 | fintech, crowdfunding, digital finance, blockchain, fintech | basic |
| 4 | global financial crisis and policies, international payments, stablec... | emerging or declining |
| 5 | financial inclusion, digital payments, india | niche |

Motor themes: digital transformation in financial services; blockchain technology applications and security.

▷ method explained: slide 40

# How the themes evolve

| **Keyword** (% of the period's papers)<br>*papers in period* | **2014–2019**<br>*136* | **2020–2022**<br>*755* | **2023–2026**<br>*2,919* | **Change** |
|---|---|---|---|---|
| financial inclusion | 5.9 | 8.2 | 15.8 | +7.6 |
| digital euro | 2.9 | 7.8 | 11.4 | +3.6 |
| global financial crisis and policies | 2.2 | 8.7 | 11.8 | +3.0 |
| financial stability | 24.3 | 12.6 | 15.6 | +3.0 |
| fintech, crowdfunding, digital finance | 48.5 | 48.7 | 49.8 | +1.0 |
| digital ruble | 0.0 | 13.9 | 14.8 | +0.9 |
| european monetary and fiscal policies | 16.9 | 10.3 | 10.3 | −0.1 |
| digital transformation in financial ser. . . | 72.1 | 74.7 | 74.2 | −0.5 |
| digital transformation in law | 10.3 | 17.9 | 17.3 | −0.6 |
| banking stability, regulation, efficien. . . | 22.1 | 16.4 | 15.5 | −0.9 |
| monetary policy and economic impact | 51.5 | 23.0 | 20.7 | −2.4 |
| digital payments | 8.8 | 12.6 | 10.2 | −2.4 |

Sorted by change between the last two periods. Darker = more frequent.

**New in each period** (c-TF-IDF, periods as classes): **2014–2019**: *e-krona*, *riksbank*, *issue*, *legal*, *innovation*, *services*; **2020–2022**: *euro*, *ruble*, *china*, *legal*, *interest*, *national*; **2023–2026**: *euro*, *legal*, *ruble*, *regulatory*, *india*, *data*.

Rising fastest: financial inclusion; falling most: digital payments.

▷ method explained: slide 45

# Topics found by meaning (SBERT + BERTopic)

| | **Topic** (top c-TF-IDF terms) | **Texts** | **% of the period's texts** | | | **Adjusted, points** | |
|---|---|---|---|---|---|---|---|
| | | | **2014–2019** | **2020–2022** | **2023–2026** | **change** | **CB** |
| 1 | cryptocurrencies, implementation, private, tr... | 470 | 18.5 | 13.5 | 8.5 | **-13.2** | **-8.6** |
| 2 | design, retail, legal, private | 439 | 10.4 | 11.2 | 8.6 | -4.0 | **-9.4** |
| 3 | uncertainty, rates, design, rate | 349 | 14.2 | 8.7 | 6.7 | **-9.6** | **-5.9** |
| 4 | dollar, trade, debt, reserve | 326 | 5.7 | 5.8 | 7.5 | +0.8 | **-5.0** |
| 5 | deposits, interest, deposit, rate | 319 | 2.8 | 5.0 | 7.3 | +3.6 | **-7.7** |
| 6 | governor, innovation, fintech, kong | 306 | 17.5 | 11.2 | 4.7 | -1.2 | **+35.7** |
| 7 | russian, russia, ruble, federation | 238 | 0.0 | 5.1 | 4.9 | **+4.7** | **-5.2** |
| 8 | china, rmb, e-cny, yuan | 213 | 1.4 | 5.3 | 4.6 | +2.7 | **-5.2** |
| 9 | cashless, evolution, riksbank, forms | 204 | 6.6 | 4.9 | 4.1 | **-3.8** | **-1.8** |
| 10 | fintech, technologies, services, innovation | 199 | 7.1 | 2.9 | 4.4 | **-3.0** | -0.5 |
| 11 | euro, european, ecb, area | 186 | 0.5 | 3.6 | 4.2 | **+3.7** | **+5.6** |
| 12 | cryptocurrencies, crypto, bitcoin, cryptocurr... | 186 | 9.0 | 4.0 | 3.6 | **-5.8** | -1.1 |

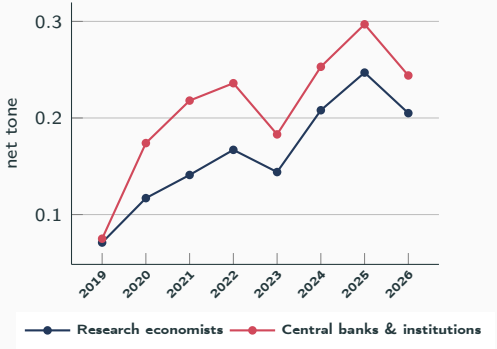
4,710 texts; 24 topics (50% outliers reassigned). Adjusted: logit on period, actor, English; change = 2023–26 vs 2014–19; CB = central banks vs research; **bold**: 95% CI excludes 0.

Topics tied to one jurisdiction: 6–34% of texts in every period, but the mix shifts: fall; *russian*, *euro*, *india* rise.

▷ method explained: slide 44

# How: tone and stance

# Tone: how positive is what economists say about CBDCs?




| | Texts | Sent. | %+ | %− | Net tone |
|---|---|---|---|---|---|
| Research economists | 2,306 | 7,519 | 21 | 5 | $0.19 \pm 0.01$ |
| Central banks & institutions | 630 | 7,576 | 18 | 3 | $0.22 \pm 0.01$ |

FinBERT on the 15,095 English sentences that name a CBDC. Sentence tone $= P(+) - P(-)$, averaged per text, then per group ($\pm$ 95% CI). 90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

Reading: tone rises steadily in both groups; research economists from 0.07 in 2019 to 0.25 in 2025; central banks stay slightly more positive.

▷ method explained: slide 48

# Do policy events move the tone?

| Date | Event | Before | After | Δtone | $p$ |
|---|---|---|---|---|---|
| Jun 2019 | Facebook announces Libra | 23 | 42 | -0.041 | 0.33 |
| Oct 2020 | ECB digital euro report | 46 | 113 | +0.003 | 0.89 |
| Oct 2020 | Bahamas launches Sand Dollar | 57 | 111 | -0.020 | 0.55 |
| Jul 2021 | ECB investigation phase | 110 | 102 | +0.020 | 0.56 |
| Oct 2021 | Nigeria launches eNaira | 115 | 103 | +0.050 | 0.24 |
| Jan 2022 | Fed CBDC discussion paper | 102 | 122 | +0.022 | 0.51 |
| Mar 2022 | US order on digital assets | 108 | 129 | +0.004 | 0.87 |
| Nov 2022 | India e-rupee pilot | 130 | 128 | +0.011 | 0.71 |
| Feb 2023 | UK digital pound consultation | 132 | 148 | -0.008 | 0.77 |
| Jun 2023 | EU digital euro bill | 145 | 211 | -0.043 | 0.30 |
| Oct 2023 | ECB preparation phase | 169 | 217 | -0.025 | 0.48 |
| Jan 2025 | US order banning a US CBDC | 193 | 230 | +0.018 | 0.60 |

For each event: mean tone of the texts in the 180 days *after* minus the 180 days *before*. Compared with 500 random dates: $p$ = share of random dates with a larger $|\Delta|$ (95th percentile: 0.084).

Result: 0 of 12 events move the tone (smallest $p = 0.24$).

Tone drifts, it does not jump: no single event explains the rise.

▷ method explained: slide 52

# Stance: for or against CBDCs?

**Level**, read by hand, % [95% CI]

| | *n* | Favourable | Critical |
|---|---|---|---|
| Research economists | 50 | 8 **[3–19]** | 2 **[0–10]** |
| Central banks & institutions | 50 | 52 **[39–65]** | 6 **[2–16]** |

By reading: 52% of central bank texts are favourable; research economists are mostly neutral (90%): they analyse, central banks advocate.

**Change** (local LLM, 575 texts sampled per actor × period), % favourable

| | 2014–2019 | 2020–2022 | 2023–2026 | Change, pts |
|---|---|---|---|---|
| Research economists | 26 | 38 | 40 | **+14** **[+1, +27]** |
| Central banks & institutions | 25 | 54 | 65 | **+40** **[+26, +53]** |

The LLM ($\kappa = 0.45$ on 100 economists' texts, best of four coders, slide 56) calls 46% of the hand-coded texts favourable against 30% by reading: its levels are inflated, so only its *changes* are used. **Bold**: CI excludes 0.

Reading: both groups become more favourable; central banks much more, as their own projects advance.

▷ method explained: slide 49

# Why: stated motivations

# Coding stated motivations

```
MOTIVES = {
  "Financial inclusion":
      r"financial\s+inclusion|unbanked|underbanked"
      r"|access\s+to\s+(?:financial\s+services|finance)",
  "Monetary sovereignty":
      r"monetary\s+sovereignty|global\s+stablecoin"
      r"|\blibra\b|big\s*tech|dollari[sz]ation",
  "Decline of cash":
      r"declin\w*\s+(?:in|of)\s+(?:the\s+)?cash"
      r"|cashless|use\s+of\s+cash",
  ...
}
```

- Motivations are stated in the introduction (else abstract, else opening).
- 10 motivations; codebook built by reading introductions; several per text allowed.
- 4,110 texts coded; 76% state at least one.
- Validated on the 50 held-out texts by economists (of 74; reading-based coding, AI-assisted): precision 0.70, recall 0.41, $\kappa = 0.46$.

▷ method explained: slide 47

# Why: what economists say CBDCs are for

| Motivation | % of texts 2014–2019 | % of texts 2020–2022 | % of texts 2023–2026 | Adjusted effect, points [95% CI] Change 2023–26 vs 2014–19 | Adjusted effect, points [95% CI] Central banks vs research |
|---|---|---|---|---|---|
| Financial stability | 30.8 | 26.3 | 29.3 | -1.6 [-8, +5] | **-7.7 [-11, -4]** |
| Monetary policy | 36.8 | 27.4 | 24.4 | **-15.1 [-22, -8]** | **-13.6 [-16, -11]** |
| Technology (DLT, programmability) | 22.5 | 21.6 | 25.8 | +1.6 [-5, +8] | **-9.1 [-12, -6]** |
| Cross-border payments | 13.2 | 19.0 | 22.5 | **+8.7 [+3, +13]** | **-7.8 [-11, -5]** |
| Payment efficiency | 10.4 | 12.1 | 17.4 | **+6.6 [+1, +11]** | **-3.1 [-6, -0]** |
| Monetary sovereignty | 15.9 | 16.8 | 13.6 | -2.0 [-8, +3] | -2.4 [-5, +0] |
| Financial inclusion | 3.8 | 8.9 | 16.8 | **+12.3 [+8, +15]** | **-9.5 [-12, -7]** |
| Geopolitics and sanctions | 2.7 | 7.2 | 15.0 | **+12.4 [+9, +15]** | +0.3 [-2, +3] |
| Privacy and surveillance | 7.1 | 10.5 | 13.6 | **+6.4 [+2, +10]** | **-2.8 [-5, -0]** |
| Decline of cash | 14.8 | 9.0 | 7.4 | **-6.5 [-12, -2]** | +1.2 [-1, +4] |

Logit of each motivation on period, actor and English; average marginal effects, robust SEs. **Bold**: CI excludes 0.

- Rising: geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts). Falling: monetary policy (−15 pts), decline of cash (−7 pts).
- From a monetary-policy instrument to a tool for inclusion, payments and geopolitics. Recall ≈ 0.41: shares are lower bounds; compare motivations and periods rather than read levels.

▷ method explained: slide 46

# Is the codebook the best coder?

| Coder | P | R | $F_1$ | $\kappa$ | $\kappa$, all |
|---|---|---|---|---|---|
| Codebook (regex) | 0.70 | 0.41 | 0.52 | 0.46 | 0.50 |
| Codebook OR classifier | 0.42 | 0.60 | 0.50 | 0.39 | 0.42 |
| SBERT, classifier (100 labels) | 0.36 | 0.37 | 0.37 | 0.25 | 0.28 |
| Local LLM, Qwen2.5 7B | 0.30 | 0.61 | 0.40 | 0.25 | 0.28 |
| Local LLM, Qwen2.5 3B | 0.31 | 0.23 | 0.26 | 0.15 | 0.16 |
| SBERT, definitions (zero-shot) | 0.21 | 0.71 | 0.32 | 0.12 | 0.15 |
| NLI zero-shot (mDeBERTa) | 0.22 | 0.39 | 0.28 | 0.11 | 0.14 |

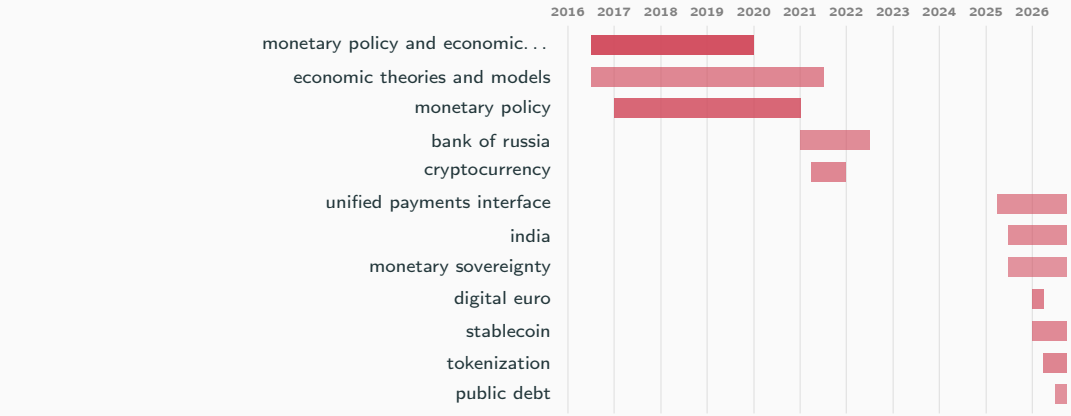
The 50 held-out texts written by economists $\times$ 10 motivations; last column: $\kappa$ on all 74 held-out texts. Thresholds and classifiers set on 100 development texts.

- The codebook wins: $\kappa = 0.46$; best alternative 0.25.
- Models confuse "mentions privacy" with "studies CBDCs because of privacy".
- Trends hold: same direction of change for 9 of 10 motivations with the high-recall coding, 9 of 10 on main-theme texts only.

▷ method explained: slide 47

# When: the environment

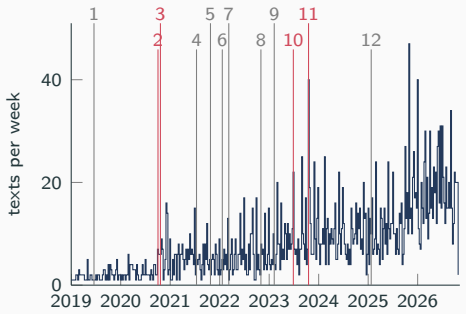
# Which themes burst, and when




Reading: early bursts on monetary policy and economic impact, economic theories and models, monetary policy; the latest on unified payments interface, india, monetary sovereignty, digital euro. ▷

method explained: slide 50

# Attention around policy events



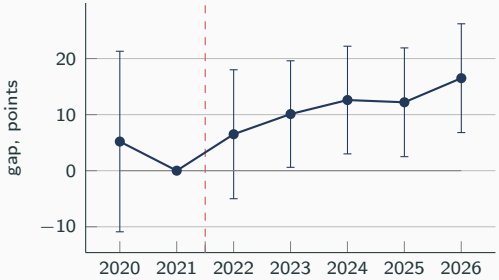

| | Date | Event | Ratio | $z$ | $p$ |
|---|---|---|---|---|---|
| 1 | Jun 2019 | Facebook announces Libra | 0.3 | -1.3 | 0.83 |
| 2 | Oct 2020 | ECB digital euro report | 2.9 | +4.0 | 0.01 |
| 3 | Oct 2020 | Bahamas launches Sand Dollar | 3.0 | +4.7 | 0.01 |
| 4 | Jul 2021 | ECB investigation phase | 0.5 | -1.9 | 0.93 |
| 5 | Oct 2021 | Nigeria launches eNaira | 0.8 | -0.7 | 0.69 |
| 6 | Jan 2022 | Fed CBDC discussion paper | 1.4 | +1.2 | 0.24 |
| 7 | Mar 2022 | US order on digital assets | 0.6 | -1.3 | 0.84 |
| 8 | Nov 2022 | India e-rupee pilot | 1.0 | -0.2 | 0.55 |
| 9 | Feb 2023 | UK digital pound consultation | 0.7 | -1.0 | 0.76 |
| 10 | Jun 2023 | EU digital euro bill | 1.7 | +2.9 | 0.05 |
| 11 | Oct 2023 | ECB preparation phase | 3.3 | +10.0 | 0.00 |
| 12 | Jan 2025 | US order banning a US CBDC | 1.2 | +0.9 | 0.29 |

Texts in the 15 days after the event vs. the rate of the 3 months before. $p$ = share of 500 random dates with a larger $z$ (the naive test flags 13% of them).

Reading: 4 of 12 events beat 95% of placebo dates; strongest: ECB preparation phase, Bahamas launches Sand Dollar.

▷ method explained: slide 51

# The 2022 sanctions: did geopolitics enter the discussion?




Gap in the share stating *geopolitics and sanctions*: texts about Russia (or in Russian) minus the others, relative to 2021; 95% CI.

756 texts about Russia among 3,772 (2020–2026). Pooled effect after 2021, every motivation:

| Motivation | DiD, points |
|---|---|
| Geopolitics and sanctions | **+11.6** [+5, +18] |
| Cross-border payments | +8.1 [-2, +18] |
| Payment efficiency | +7.1 [-1, +15] |
| Decline of cash | +2.8 [-2, +8] |
| Privacy and surveillance | -4.0 [-11, +3] |
| Financial inclusion | **-6.7** [-13, -0] |
| Monetary sovereignty | -8.7 [-18, +1] |
| Technology (DLT, programmability) | -10.0 [-21, +1] |
| Monetary policy | **-12.0** [-24, -0] |
| Financial stability | **-12.6** [-25, -1] |

Reading: geopolitics rises by +11.6 points [+5, +18] more in texts about Russia, and the gap widens each year. But 3 other motivations shift too: consistent with a sanctions effect, not proof.

▷ method explained: slide 53

# The discussion and its environment

# Putting it together, period by period

| | 2014–2019 | 2020–2022 | 2023–2026 |
|---|---|---|---|
| **Policy events (bold: attention reacts)** | **Facebook announces Libra** | ECB digital euro report; Bahamas launches Sand Dollar; **ECB investigation phase**; **Nigeria launches eNaira**; **Fed CBDC discussion paper**; **US order on digital assets**; India e-rupee pilot | **UK digital pound consultation**; EU digital euro bill; ECB preparation phase; **US order banning a US CBDC** |
| **Texts per year** | **53** | **324** | **840** |
| **Bursting keywords** | **monetary policy and economic impact, economic theories and models, monetary policy** | **bank of russia, cryptocurrency** | **unified payments interface, india, monetary sovereignty** |
| **Most frequent motivations** | **monetary policy, financial stability, technology** | **monetary policy, financial stability, technology** | **financial stability, technology, monetary policy** |
| **Net tone (research / central banks)** | **0.06 / 0.10** | **0.15 / 0.22** | **0.21 / 0.24** |
| **Favourable, LLM (research / central banks)** | **26% / 25%** | **38% / 54%** | **40% / 65%** |

- 2014–19: a monetary-policy question, alongside Libra and the decline of cash; few texts, cautious tone.
- 2020–22: central bank reports and pilots; the discussion grows and turns favourable.
- 2023–26: national projects, sanctions, cross-border payments; the most favourable period.

# What we can and cannot say about causes

| Link | Test | Result | Verdict |
|---|---|---|---|
| Events → attention | event study vs. placebo dates | 4 of 12 events | **timing evidence** |
| Events → tone | 180-day windows vs. placebo dates | 0 of 12 events | **no effect found** |
| Sanctions → geopolitics motive | difference-in-differences | +11.6 points [+5, +18] | **consistent with** |
| Events → stance | – | sample too small around each date | not testable |
| Motivations ↔ stance | – | measured in the same texts | association only |

- The paper stays descriptive: it measures changes and shows what moves with them.
- Explaining *why* economists changed their stance would need other evidence (interviews, authors' careers): hypotheses for further work.

# Robustness

# Robustness

| Specification | Texts | Why: change, points inclusion | Why: change, points monetary policy | How: tone CB − research | How: stance fav. CB − research | When events |
|---|---|---|---|---|---|---|
| Baseline: economists | 4,710 | +12.3 [+8, +15] | −15.1 [-22, -8] | +0.03 | +16 | 4/12 |
| Main theme only | 3,513 | +14.9 [+10, +18] | −16.0 [-25, -7] | +0.01 | +13 | 4/12 |
| English only | 3,721 | +13.7 [+9, +16] | −16.0 [-23, -7] | +0.03 | +16 | 4/12 |
| Journal articles only (research) | 3,171 | +12.3 [+6, +15] | −16.8 [-25, -8] | +0.01 | +11 | 4/12 |
| Without 2026 (partial year) | 3,547 | +11.8 [+7, +15] | −15.3 [-23, -8] | +0.03 | +15 | 4/12 |
| High-recall motivation coding | 4,710 | +15.6 [+10, +19] | −8.8 [-16, -2] | +0.03 | +16 | 4/12 |
| All disciplines (not only economists) | 8,213 | +13.2 [+10, +15] | −15.0 [-21, -9] | +0.02 | +15 | 4/12 |

Changes: 2023–26 vs 2014–19, adjusted (logit, 95% CI). Tone: mean net tone gap. Stance: raw LLM labels, points. Events: $p < 0.05$ against placebo dates.

Signs and orders of magnitude hold in every specification, including when research from every discipline is kept.

▷ method explained: slide 54

# Conclusion

# Takeaways and limits

**How economists' discussion of CBDCs evolved**

- Who: 4,710 texts; central bank economists write 10% of the papers but get 48% of citations.
- What: from monetary policy and design to inclusion and national projects (digital ruble, rupee, e-CNY).
- How: tone and stance become more favourable; research economists stay mostly neutral, central banks advocate.
- Why: rising geopolitics and sanctions (+12 pts), financial inclusion (+12 pts), cross-border payments (+9 pts); falling monetary policy (−15 pts), decline of cash (−7 pts).
- When: 4 of 12 events move attention, 0 of 12 move tone; geopolitics rises after the 2022 sanctions.

**Limits**

- Economists identified by field and affiliation, not by training.
- Motivations: recall $\approx 0.5$ (lower bounds). Tone: English only. Stance: LLM for changes only.
- One coder (AI-assisted) for the validation samples.
- Descriptive: causes stay hypotheses.

# Methods explained

# Methods explained: how to read this appendix

One slide per tool, each in four parts:

- **Idea**: what the tool does, in words.
- **How**: the computation, step by step, with its formula.
- **Why this tool**: what it solves, and what it was compared with.
- **Used on**: links to the result slides.


| Tool | Slide | Tool | Slide |
|---|---|---|---|
| Who is an economist | slide 38 | Covariate-adjusted logit | slide 46 |
| Networks: association strength, Louvain | slide 39 | Codebook and validation | slide 47 |
| Thematic map (Callon) | slide 40 | Tone (FinBERT) | slide 48 |
| Bibliometric laws | slide 41 | Stance (reading + LLM) | slide 49 |
| Log-odds with Dirichlet prior | slide 42 | Bursts (Kleinberg) | slide 50 |
| Sentence embeddings (SBERT) | slide 43 | Event study and placebo | slide 51 |
| Topic model (BERTopic) | slide 44 | Tone around events | slide 52 |
| Class-based TF-IDF | slide 45 | Difference-in-differences | slide 53 |
| | | Robustness | slide 54 |

# Method: who counts as an economist

**Idea**

The course asks how *economists* discuss CBDCs, so each text must be traced to an economist or an economic institution.

**How**

- **Research**: kept if OpenAlex classifies it in *Economics, Econometrics and Finance* (primary topic, from title, abstract, venue, citations), or an author works at a central bank, the IMF or the BIS; without OpenAlex record: an economics venue.
- **Institutions**: central bank speeches; documents of central banks and international financial institutions.
- **Dropped**: other fields (computer science, law. . . ), governments and parliaments, the press.

**Why this tool**

Authors' training is not recorded anywhere at scale; the field of the paper and the employer of the author are, and they are what defines a discipline in bibliometrics.

Speeches are kept as *institutional* voices: written with the bank's economics staff, even when the speaker is not an economist.

Robustness: results with every discipline kept are on slide 35.

**Used on**

slide 12, and every result after it.

# Method: networks (association strength, Louvain)

**Idea**

Two items (references, authors, keywords) are linked when they appear in the same paper; groups of tightly linked items are schools, teams or themes.

**How**

1. Count $c_{ij}$ = papers where $i$ and $j$ appear together; $c_i$ = papers with $i$.
2. Normalise by association strength $s_{ij} = c_{ij}/(c_i c_j)$: the ratio of observed to expected co-occurrence.
3. Cluster with Louvain: move nodes between groups while the modularity $Q = \frac{1}{2m}\sum_{ij}\left[s_{ij} - \frac{k_i k_j}{2m}\right]\delta(g_i, g_j)$ rises.
4. Draw: force-directed layout; size $\propto \sqrt{c_i}$.

**Why this tool**

Raw counts favour popular items: two classics are co-cited often by chance. Association strength removes this (van Eck & Waltman, 2009); it is VOSviewer's default.

Louvain is fast and needs no number of clusters; $Q$ tells how clear the grouping is.

**Co-citation** (references cited together) shows the intellectual base; **co-authorship** shows teams.

**Used on**

slide 18, slide 19, slide 21.

# Method: thematic map (Callon)

**Idea**

Place each cluster of keywords on two axes: how much it is linked to the rest of the field, and how developed it is inside.

**How**

With the equivalence index $e_{ij} = c_{ij}^2/(c_i c_j)$:

$$\text{centrality} = 10 \sum_{i \in C,\, j \notin C} e_{ij}, \qquad \text{density} = \frac{100}{|C|} \sum_{i,j \in C} e_{ij}$$

Quadrants, split at the medians:

- **motor**: central and dense; **basic**: central, loose;
- **niche**: dense, isolated; **emerging or declining**: neither.

**Why this tool**

A keyword list says what is frequent; the map says what *structures* the field. It is the standard tool of science mapping (Callon et al. 1991; Cobo et al. 2011; bibliometrix).

Limit: keywords are partly machine-assigned, so the topic model (meaning-based) is used alongside.

**Used on**

slide 21.

# Method: bibliometric laws

**Idea**

Three regularities describe the shape of any scientific field: how fast it grows, how output is spread among authors, and among journals.

**How**

- **Growth**: compound annual rate $g = (N_{y_1}/N_{y_0})^{1/(y_1 - y_0)} - 1$.
- **Lotka**: share of authors with $n$ papers $f(n) \propto n^{-\beta}$; $\beta$ fitted on the log-log counts ($\beta \approx 2$ in mature fields).
- **Bradford**: sort sources by output, cut in three zones holding $1/3$ of papers each; the first zone is the core.

**Why this tool**

They situate the field: a high growth rate and a large $\beta$ (most authors write once) mean a young, open field around a small core of outlets.

They are the descriptive baseline of every bibliometric review, so our numbers can be compared with the earlier reviews.

**Used on**

slide 14, slide 16.

# Method: distinctive words (log-odds with Dirichlet prior)

**Idea**

Which words does one group use *more* than the other, once chance is accounted for?

**How**

For word $w$, group $i$ vs. the other group $j$, with counts $y$, totals $n$, and a prior $\alpha_w = \alpha_0 \times$ (corpus frequency of $w$):

$$\delta_w = \log \frac{y_{iw} + \alpha_w}{n_i + \alpha_0 - y_{iw} - \alpha_w} - \log \frac{y_{jw} + \alpha_w}{n_j + \alpha_0 - y_{jw} - \alpha_w}$$

$$z_w = \delta_w \Big/ \sqrt{\frac{1}{y_{iw} + \alpha_w} + \frac{1}{y_{jw} + \alpha_w}}$$

Words are ranked by $z_w$.

**Why this tool**

Plain frequencies list words everyone uses; TF-IDF or raw log-ratios are dominated by rare words (a word used twice by one group gets an infinite ratio). The prior shrinks rare words towards zero and the $z$-score accounts for uncertainty (Monroe, Colaresi & Quinn, 2008).

**Used on**

slide 20.

# Method: sentence embeddings (SBERT)

**Idea**

Turn each text into a vector of 384 numbers such that texts with similar *meaning* get close vectors, whatever their words or language.

**How**

- A transformer network reads title + abstract (or opening) and averages its word representations into one vector $\mathbf{e}_t$.
- Model: `paraphrase-multilingual-MiniLM-L12-v2`, trained so that translations and paraphrases land together (50+ languages).
- Similarity: cosine $\cos(\mathbf{e}_s, \mathbf{e}_t) = \frac{\mathbf{e}_s \cdot \mathbf{e}_t}{\|\mathbf{e}_s\| \|\mathbf{e}_t\|}$.

**Why this tool**

Keywords and word counts treat "digital ruble" and its Russian original (*tsifrovoy rubl'*) as unrelated; embeddings do not. With 79% English and many Russian texts, this is what makes a single topic model possible.

Tested also as a motivation coder (similarity to prototypes): it lost to the codebook (slide 29).

**Used on**

slide 23.

# Method: topic model (BERTopic)

**Idea**

Find groups of texts about the same subject, without fixing the subjects in advance.

**How**

1. **Embed** each text (SBERT).
2. **Reduce** 384 dimensions to 5 with UMAP, which keeps neighbours together.
3. **Cluster** with HDBSCAN: dense regions become topics (at least 40 texts; leaf selection, which favours fine topics over one large blob); texts in no dense region are outliers, then assigned to the nearest topic.
4. **Label** each topic by its c-TF-IDF terms (slide 45).

**Why this tool**

LDA and STM count words, one language at a time, and need the number of topics. BERTopic works on meaning, across languages, and finds the number itself (Grootendorst, 2022).

The covariate part of STM is kept: topic shares are adjusted for period, actor and language (slide 46).

**Used on**

slide 23.

# Method: class-based TF-IDF (c-TF-IDF)

**Idea**

Which words characterise a class of texts (a topic, or a period) compared with all the others?

**How**

Merge the texts of each class $c$ into one document; for term $t$:

$$W_{t,c} = \frac{\mathrm{tf}_{t,c}}{|c|} \log\left(1 + \frac{A}{f_t}\right)$$

$\mathrm{tf}_{t,c}$ = count of $t$ in class $c$, $|c|$ = words in $c$, $A$ = average words per class, $f_t$ = count of $t$ in all classes. Words in more than 15% of texts and generic words are removed.

**Why this tool**

Ordinary TF-IDF scores words within one document; here the unit is a whole class. A frequent word in one class but rare elsewhere scores high: it is what is *new* or *specific*.

Periods as classes answer "when": which vocabulary appears in each period.

**Used on**

slide 23 (topic labels), slide 22 (periods).

# Method: covariate-adjusted logit

**Idea**

A raw change between periods can come from a change in *who* writes (more central banks, more Russian texts). Compare like with like.

**How**

For each outcome $y_t \in \{0, 1\}$ (a motivation, a topic):

$$P(y_t = 1) = \Lambda(a + b_{\text{period}} + c_{\text{actor}} + e \cdot \text{English})$$

- **Average marginal effect**: predict every text in 2023–26, then in 2014–19, average the difference (points).
- 95% CI: robust (HC1) standard errors; coefficients drawn 400 times from their estimated distribution.

**Why this tool**

It is the "prevalence" model of a structural topic model (Roberts et al., 2014), applied to any coded outcome. Marginal effects are read in percentage points, unlike log-odds coefficients.

**Used on**

slide 28, slide 23, slide 35.

# Method: codebook and validation

**Idea**

Write down, for each of 10 motivations, the phrases authors use to state it; flag a text when its introduction contains one.

**How**

1. Read 100 introductions (**development set**); write regular expressions in 7 languages.
2. Code 74 *other* texts by reading (**test set**), never used for tuning; scores on the 50 written by economists.
3. Compare: precision $= \frac{TP}{TP+FP}$, recall $= \frac{TP}{TP+FN}$, Cohen's $\kappa = \frac{p_o - p_e}{1-p_e}$ (agreement beyond chance).

**Why this tool**

Motivations need reading; a codebook makes the reading explicit, reproducible and checkable.

Five machine coders were tested on the same texts (SBERT $\times$ 2, NLI mDeBERTa, Qwen2.5 3B and 7B): see slide 29.

High precision, recall $\approx 0.5$: shares are lower bounds; changes are robust (slide 29).

**Used on**

slide 27, slide 28, slide 29.

# Method: tone (FinBERT)

**Idea**

Measure how positive or negative the sentences *about CBDCs* are.

**How**

1. Keep the English sentences that name a CBDC (same term list as the filter).
2. FinBERT (a BERT model trained on financial news) gives $P(+), P(0), P(-)$ for each sentence.
3. Sentence tone $= P(+) - P(-) \in [-1, 1]$; averaged per text, then per group and year.
4. Validation: 90 sentences from economists' texts read blind; $\kappa = 0.52$ (all 0.53).

**Why this tool**

Dictionary methods (Loughran–McDonald) miss context ("not risky"); general sentiment models misread finance ("liability" is not negative). FinBERT is the standard for economic text (Araci, 2019).

Scoring only CBDC sentences isolates the tone *about CBDCs*, not about the economy.

Limit: it sees tone in neutral sentences, so compare groups and years, not levels.

**Used on**

slide 24, slide 25.

# Method: stance (reading + local language model)

**Idea**

Tone is about words; stance is about the *author's position*: favourable, neutral or critical towards issuing a CBDC.

**How**

- **Levels**: texts read by hand per actor; share with a Wilson interval $\frac{\hat{p}+\frac{z^2}{2n} \pm z\sqrt{\frac{\hat{p}(1-\hat{p})}{n}+\frac{z^2}{4n^2}}}{1+z^2/n}$.
- **Changes**: Qwen2.5 7B, run locally, labels a random sample of up to 100 texts per actor × period; change $= \hat{p}_{\text{late}} - \hat{p}_{\text{early}}$, CI $\pm 1.96\sqrt{\sum \hat{p}(1-\hat{p})/n}$.

**Why this tool**

Stance needs understanding of the argument: only a language model can do it at scale. Four coders were tested (slide 56); the LLM is best ($\kappa = 0.45$) but over-calls "favourable".

A constant bias cancels in a difference: so the LLM gives *changes*, reading gives *levels*.

**Used on**

slide 26, slide 33.

# Method: bursts (Kleinberg)

**Idea**

Let the data say when a keyword suddenly becomes frequent, instead of choosing dates.

**How**

For keyword $w$, quarter $t$: $r_t$ of $d_t$ papers use it.

- Two states: normal rate $p_0 = R/D$ (overall), burst rate $p_1 = 2p_0$.
- Cost of a quarter in state $i$: $\sigma_i(t) = -\ln\left[\binom{d_t}{r_t} p_i^{r_t}(1-p_i)^{d_t - r_t}\right]$; entering a burst costs $\ln T$.
- The cheapest sequence of states (Viterbi algorithm) gives the bursts; weight = cost saved.

**Why this tool**

A simple growth rate reacts to noise; the entry cost means only sustained rises count. It is the standard "burst detection" of CiteSpace and science-of-science work (Kleinberg, 2002).

**Used on**

slide 30, slide 33.

# Method: event study with placebo dates

**Idea**

Does writing on CBDCs jump right after a policy event, more than on an ordinary day?

**How**

- Baseline rate $\lambda$ = texts per day over days $[-90, -15]$ before the event.
- Expected over $[0, 14]$: $E = 15\lambda$; ratio $= N/E$; $z = (N - E)/\sqrt{E}$.
- Placebo: the same $z$ on 500 random dates; $p$ = share of random dates with a larger $z$.

**Why this tool**

The usual Poisson test assumes a stable rate; attention trends upward and comes in waves, so 13% of random dates look "significant". Comparing with random dates gives the right yardstick (as in finance event studies, MacKinlay 1997).

It is causal only in a narrow sense: timing of attention.

**Used on**

slide 31, slide 34.

# Method: do events move the tone?

**Idea**

Same logic as the event study, applied to the tone of the texts rather than their number.

**How**

- Texts dated to the day, with a tone score.
- $\Delta$ = mean tone over $[t_0, t_0 + 180)$ minus mean tone over $[t_0 - 180, t_0)$.
- Same $\Delta$ on 500 random dates; $p$ = share with $|\Delta|$ at least as large (two-sided).

**Why this tool**

Tone rises over time anyway; a before/after difference alone would credit every event with that trend. Random dates carry the same trend, so only a jump beyond it counts.

180 days: enough texts on each side (100+).

**Used on**

slide 25, slide 34.

# Method: difference-in-differences

**Idea**

If the 2022 sanctions pushed economists to discuss geopolitics, texts about Russia should change *more* than the others after 2022.

**How**

Linear probability model, robust SEs; $y$ = motivation stated, $T$ = text about Russia or in Russian:

$$y_{it} = a_T + b_t + \sum_{k \neq 2021} \gamma_k \, T_i \, 1[t = k] + \varepsilon_{it}$$

$\gamma_k$ = gap in year $k$ relative to 2021. Pooled: one $\gamma$ for $t > 2021$.

**Why this tool**

A simple before/after mixes the sanctions with everything else that happened in 2022; the other texts serve as the control.

Two checks: **pre-trends** ($\gamma_k \approx 0$ before 2022: only 2020 is usable) and **placebo outcomes** (other motivations should not move: some do). Hence "consistent with", not proof.

**Used on**

slide 32, slide 34.

# Method: robustness

**Idea**

Re-run the headline results on samples that remove one possible source of error at a time.

**How**

- **Main theme only**: the secondary theme label is noisy.
- **English only**: language mix changes over time.
- **Journal articles only**: working papers and chapters differ.
- **Without the current year**: partial year.
- **High-recall coding**: codebook + SBERT matches.
- **All disciplines**: the economist filter itself.

**Why this tool**

Each measure has a known weakness; a result that holds whichever way the sample is cut does not depend on it. It answers a referee's "what if" in one table.

**Used on**

slide 35.

# Backup

# Backup: validation of the motivation coding

Codebook tuned on 100 texts (development set), then evaluated on the 50 held-out texts written by economists (of 74); unit = first 1,500 characters of the introduction / abstract.

| Motivation | Read | Coded | Precision | Recall | $\kappa$ |
|---|---|---|---|---|---|
| Privacy and surveillance | 3 | 2 | 1.00 | 0.67 | 0.79 |
| Geopolitics and sanctions | 7 | 4 | 1.00 | 0.57 | 0.70 |
| Technology (DLT, programmability) | 11 | 11 | 0.73 | 0.73 | 0.65 |
| Financial inclusion | 7 | 3 | 1.00 | 0.43 | 0.56 |
| Monetary policy | 7 | 9 | 0.56 | 0.71 | 0.55 |
| Cross-border payments | 1 | 3 | 0.33 | 1.00 | 0.48 |
| Financial stability | 10 | 7 | 0.57 | 0.40 | 0.37 |
| Monetary sovereignty | 6 | 1 | 1.00 | 0.17 | 0.26 |
| Decline of cash | 7 | 1 | 1.00 | 0.14 | 0.22 |
| Payment efficiency | 16 | 3 | 0.67 | 0.12 | 0.12 |
| **All** | 75 | 44 | 0.70 | 0.41 | 0.46 |

Read = texts where the reading found the motivation; Coded = texts flagged by the codebook. Files: `presentation/validation_motivations_dev.csv`, `_test.csv`.

# Backup: validation of the stance coders

Each coder against the stance read by hand (favourable / neutral / critical); thresholds and classifiers by 5-fold cross-validation.

| **Coder** | **Accuracy** | $\kappa$ | $\kappa$, **all** |
|---|---|---|---|
| Local LLM, Qwen2.5 7B | 0.70 | 0.45 | 0.41 |
| FinBERT tone | 0.65 | 0.31 | 0.20 |
| NLI zero-shot (mDeBERTa) | 0.63 | 0.29 | 0.18 |
| SBERT classifier | 0.61 | 0.21 | 0.17 |

Scores on the 100 texts written by economists; last column: all 177 hand-coded texts (including the press).

Files: `presentation/validation_stance.csv`, `validation_predictions_stance.csv`.

# Backup: validation of the tone measure

Sentences sampled per FinBERT label, read blind to the label; table: the 90 from economists' texts (30 per label).

| FinBERT ↓ / reading → | negative | neutral | positive | Precision |
|---|---|---|---|---|
| negative | 17 | 11 | 2 | 0.57 |
| neutral | 2 | 25 | 3 | 0.83 |
| positive | 1 | 10 | 19 | 0.63 |

90 sentences from economists' texts read blind (of 136): agreement 68%, $\kappa = 0.52$; positive and negative confused 3 times.

File: `presentation/validation_tone.csv`.

# Backup: how the relevance score behaves

| Example | Score | Label |
|---|---|---|
| Paper, CBDC in title | 87 | main theme |
| Paper, generic title, CBDC abstract | 80 | main theme |
| Paper, one passing mention | 11 | mentioned only |
| Speech, one paragraph on it | 18 | mentioned only |
| Speech devoted to it | 80 | main theme |
| Medical case report | 0 | off-topic |

# Backup: press, recursive bisection of time windows

```
stack = monthly_windows("2000-01-01", today)
while stack:
    a, b = stack.pop()
    items = fetch(query, after=a, before=b)
    if len(items) >= CAP and (b - a).days > 1:
        m = a + (b - a) / 2
        stack += [(m, b), (a, m)]
        continue
    save(items)
    checkpoint(query, a, b, len(items))
```

A news search returns at most $C = 100$ items. A window holding more is cut in two until each holds fewer, down to one day: depth $\leq 5$ per month, $2L - 1$ requests for $L$ windows. The press was collected, then left out of the economists' corpus.

# Backup: academic sources, cursor pagination under a rate limit

```
def iter_phrase(phrase):
    cursor = "*"
    while cursor:
        d = get(f'{API}?filter=title_and_abstract'
                f'.search:"{phrase}"&per-page=200'
                f'&cursor={cursor}')
        yield from d["results"]
        cursor = d["meta"].get("next_cursor")

class Throttle:
    def wait(self):
        dt = self.interval - (now() - self.last)
        if dt > 0: sleep(dt)
        self.last = now()
```

- Each phrase searched in titles and abstracts; results merged on the source id.
- Requests spaced by $t_{k+1} - t_k \geq \Delta$.
- On HTTP 429: back-off $w_j = \min(60j,\, 900)$ s.
- Reference lists resolved into citations (for co-citation).

# Backup: full text and de-duplication

**Full text**

- PDFs (PyMuPDF) and web pages (trafilatura).
- Split into introduction / conclusion / section headings.

```
def key(r):
    if r.source_type == "academic":
        if r.doi: return "doi:" + r.doi.lower()
        return f"t:{norm(r.title)[:120]}:{r.year}"
    if r.source_type == "speech":
        return f"s:{r.date}:{norm(r.title)[:50]}"
    return "u:" + r.url
```

Merge: most precise date, longest abstract, union of keywords.

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